Adrien Vacher
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EDUCATION

INRIA and ENSAE, PARIS 2023 –
Postdoc: Non-log-concave sampling and diffusion models.
Advisors: Flavien Léger and Anna Korba.

UNIVERSITÉ GUSTAVE EIFFEL, PARIS 2020 – 2023
PhD in Machine-Learning: numerical methods for regular optimal transport.
Advisor: François-Xavier Vialard.

ECOLE NORMALE SUPÉRIEURE, SACLAY 2018 – 2019
Master Vision et Apprentissage (MVA).

ENSAE, PARIS 2016 – 2019
B.S. and M.S. in Applied Mathematics.

LYCÉE SAINT-LOUIS, PARIS 2013 – 2016
Classe préparatoire filière MP.

WORK EXPERIENCE

NYU, PARIS 2023 - 2026
Teacher assistant in applied mathematics.

ENSAE, PARIS 2023 - 2025
Teacher assistant in applied mathematics.

UNIVERSITÉ GUSTAVE EIFFEL, PARIS 2020 - 2023
Teacher assistant in applied mathematics.

DEEZER R&D, PARIS 2019
Research Internship (6 months): large-scale recommender systems.

JABBA TECHNOLOGIES, SINGAPORE 2018
Internship (6 months): data scraping and statistical arbitrage of cryptocurrencies.

PUBLICATIONS

- G. Menon, A. Stromme, A. Vacher, *On the implicit regularization of Langevin dynamics with projected noise,* COLT 2026.
- A. Vacher, O. Chehab, A. Korba, *Sampling from multimodal distributions in polynomial time in fixed dimension via reverse diffusion,* NeurIPS 2025.
- O. Chehab, A. Korba, A. Stromme, A. Vacher, *Provable convergence and limitations of Geometric Tempering for Langevin Dynamics,* ICLR 2025.
- A. Vacher, B. Muzellec, F. Bach, F.-X. Vialard and A. Rudi, *Minimax Estimation of smooth Optimal Transport maps with kernel Sum-Of-Squares*, SIAM Journal on Mathematics of Data Science, SIMODS 2024.
- A. Vacher and F.-X. Vialard, *Semi-dual Unbalanced Optimal Transport,* ICML 2023.
- A. Vacher and F.-X. Vialard, *Model selection in Optimal Transport via Brenier Semi-Dual*, NeurIPS 2022.
- A. Vacher, B. Muzellec, A. Rudi, F. Bach and F.-X. Vialard, *A Dimension Free Computational Upper Bound for Smooth Optimal Transport Estimation*, COLT 2021.

LANGUAGES & TECHNOLOGY

- Python, TensorFlow, PyTorch, MOSEK, Git, LaTex.
- **French**: native, **English**: fluent (TOEFL 109/120), **Spanish**: conversational, **Hebrew**: basics.